

Project proposal

The Islamia University of Bahawalpur

Faculty of computing

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project proposal

Project Title

Brain tumor detection using Convolutional Neural Network (CNN)

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1. Introduction

Brain tumors are one of the most serious health conditions that require early and accurate diagnosis. Traditionally, doctors analyze MRI (Magnetic Resonance Imaging) scans manually, which can be time-consuming and prone to human error.

With the advancement of Machine Learning and Deep Learning, automated systems can assist in detecting tumors more efficiently and accurately. This project aims to develop an intelligent system that can analyze MRI images and detect the presence of brain tumors.

2. Problem Statement

Manual analysis of brain MRI scans is:

Time-consuming

Dependent on expert radiologists

Prone to human error

There is a need for an automated system that can:

Quickly analyze MRI images

Accurately detect tumors

Assist medical professionals in diagnosis

3. Objectives

The main objectives of this project are:

To develop a machine learning model for brain tumor detection

To classify MRI images into tumor and non-tumor categories

To achieve high accuracy using deep learning techniques

To design a simple user interface for image upload and prediction

4. Methodology

Step 1: Data Collection

Collect MRI brain image dataset from reliable sources (e.g., Kaggle)

Step 2: Data Preprocessing

Resize images

Normalize pixel values

Label data (tumor / no tumor)

Step 3: Model Development

Use Convolutional Neural Network (CNN)

Train model on dataset

Step 4: Model Evaluation

Measure accuracy, loss, and performance

Use validation techniques

Step 5: Deployment

Develop a simple web application using Streamlit

Allow users to upload MRI images and get predictions

5. Tools & Technologies

Programming Language: Python

Libraries: TensorFlow, Keras, OpenCV, NumPy

Framework: Streamlit

Platform: Jupyter Notebook / VS Code

6. Expected Outcomes

A trained CNN model capable of detecting brain tumors

Accuracy of approximately 85%–95%

A functional web-based interface

Fast and reliable predictions

7. Significance of the Project

This project can:

Assist doctors in early diagnosis

Reduce workload of radiologists

Improve accuracy and efficiency

Serve as a foundation for future medical AI systems

8. Limitations

Depends on dataset quality

Not a replacement for professional medical diagnosis

Requires computational resources for training

9. Future Enhancements

Tumor classification (benign vs malignant)

Tumor localization (highlight area)

Integration with hospital systems

10. Conclusion

This project proposes an AI-based solution for detecting brain tumors from MRI images using deep learning techniques. The system aims to provide accurate and fast predictions, supporting medical professionals in diagnosis and improving healthcare efficiency.